Akoizo Original Practice - 2027 Rotation Form 1

Division B | Remote Sensing | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Classify Sentinel-1 SAR as active or passive and explain the distinction.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

2. (2 points) Which kind of resolution describes the ground area represented by a pixel?

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

3. (2 points) Does a shorter orbital period necessarily give an equally short revisit time for a specified ground point? Explain.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

4. (2 points) Using c=3.0e8 m/s, find the frequency of light with wavelength 6.0e-7 m.

Answer: _____
_____
_____

5. (2 points) Use Wien's law with b=2.90e-3 m K to find peak wavelength at 290 K.

Answer: _____
_____
_____

6. (2 points) A radar echo takes 2.0e-5 s for its round trip. With c=3.0e8 m/s, calculate range.

Answer: _____
_____
_____

7. (2 points) Find the changed patch east-west ground length.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

8. (2 points) Find the changed patch north-south ground length.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

9. (2 points) Find its ground area.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

10. (2 points) Find its perimeter.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

11. (2 points) Cloud obscures the optical image but the SAR image shows a surface pattern. Explain why these observations differ.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

12. (2 points) A dark SAR patch is labeled water from brightness alone. Give one alternative explanation and a stronger check.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

Answer: _____
_____
_____

13. (2 points) A classified map labels 41 of 100 equal-area pixels as forest. Calculate forest cover percentage.

Answer: _____
_____
_____

14. (2 points) Forest pixels decline from 81 to 51. How many equal-area pixels of forest are lost?

Answer: _____
_____
_____

15. (2 points) Explain why comparing a wet-season image with a dry-season image can mimic land-cover change.

Answer: _____
_____
_____

16. (2 points) A topographic profile rises 60 m over 1,200 m horizontal distance. Find gradient as a percentage.

Answer: _____
_____
_____

17. (2 points) Two contours are 20 m apart vertically. Adjacent contours are closer horizontally in area A than area B. Which area is steeper and why?

Answer: _____
_____
_____

18. (2 points) A false-color display makes healthy vegetation red. Does that mean the vegetation reflects red light most strongly? Explain.

Answer: _____
_____
_____

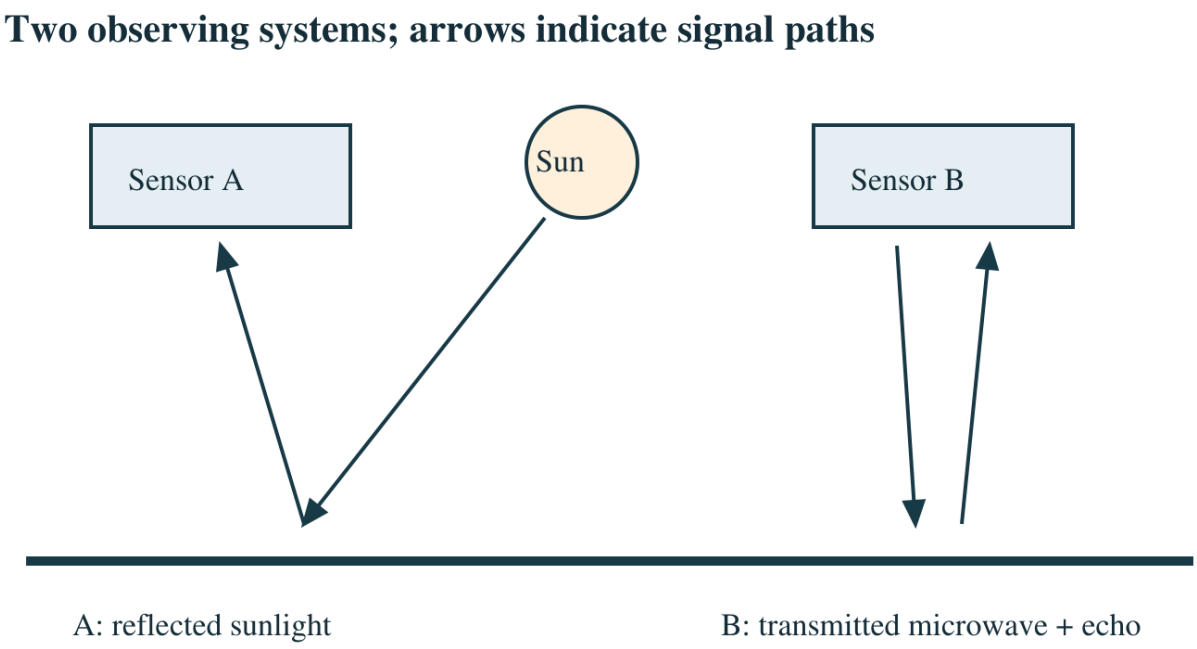
19. (2 points) An ECOSTRESS thermal image shows a warmer paved area than a nearby canopy. Give an observation and a cautious interpretation.

Answer: _____
_____
_____

20. (2 points) A classifier was trained using labeled examples of water, forest and pavement. Is it supervised or unsupervised? Explain.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Classify A as active or passive using the arrows.



F1. Active and passive observing geometries (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Classify B as active or passive using the arrows.

Refer to F1. Active and passive observing geometries (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

23. (2 points) Use Figure F1. Which path is generally better suited to observing the surface through cloud, and why?

Refer to F1. Active and passive observing geometries (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Which resolution category is quantified by the pixel side length?

Synthetic classified maps, not real mission imagery

Landsat: earlier

F	F	F	F	W	W
F	F	F	F	W	W
B	B	B	B	W	W
B	B	B	B	W	W

Landsat: later

F	F	B	B	W	W
F	F	B	B	W	W
B	B	B	B	W	W
B	B	B	B	W	W

F forest; B built; W water. Same grid, north up.

Each square pixel = 15 m x 15 m. Dates matched by season.

F2. Paired classified Landsat maps (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. How many forest pixels change to built class?

Refer to F2. Paired classified Landsat maps (original synthetic teaching figure) shown with the first question in this figure group.

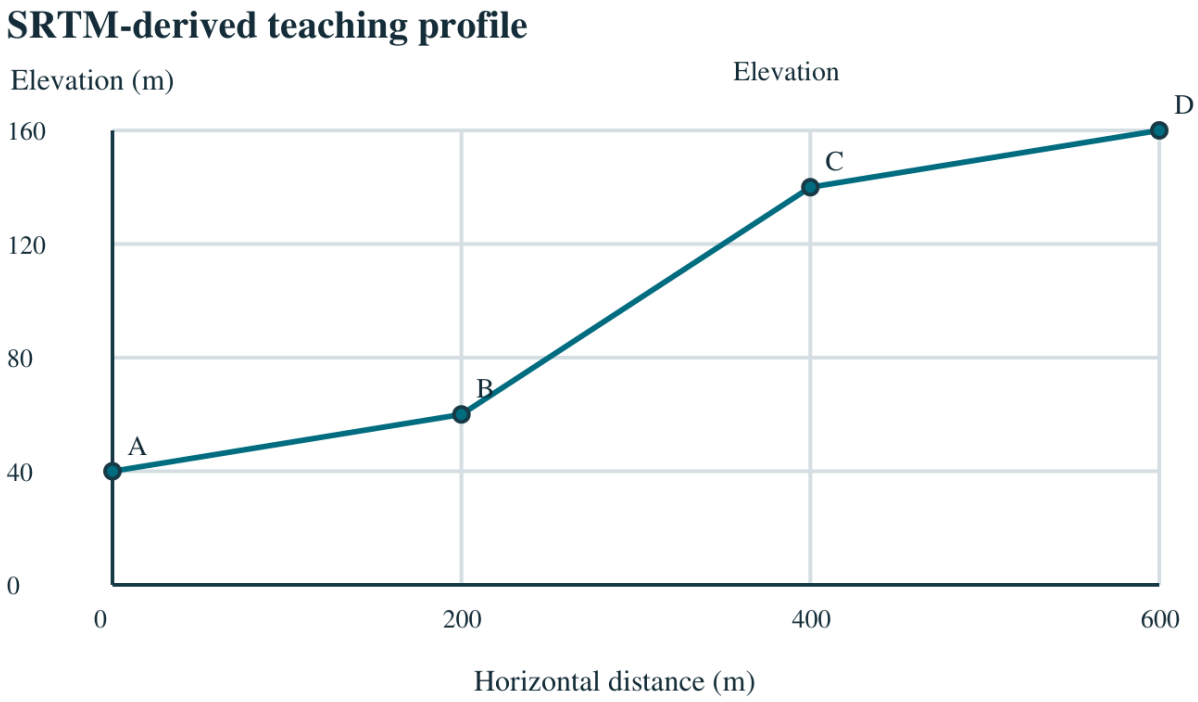
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Calculate the ground area of the forest-to-built change.

Refer to F2. Paired classified Landsat maps (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Calculate mean slope from B to C as a percentage.



F3. Elevation transect (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Which labeled interval is steepest? Justify from the axes.

Refer to F3. Elevation transect (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

29. (2 points) Use Figure F3. Read the total elevation difference between A and D.

Refer to F3. Elevation transect (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Calculate pavement-minus-canopy surface temperature.

Synthetic ECOSTRESS/Landsat interpretation

Feature	Thermal surface (C)	NIR display channel	Reference class
A	38	bright red	Canopy
B	48	dark red	Pavement
C	25	very dark	Open water

F4. Thermal and false-color products (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Explain why bright red A does not mean its leaves reflect visible red most strongly.

Refer to F4. Thermal and false-color products (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Can the thermal panel alone establish a 10 C air-temperature difference between A and B? Explain.

Refer to F4. Thermal and false-color products (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____